Step 9 Block-Structure, Order, and Adaptation Analysis

Overview

This report implements the Step 9 block-adjusted analysis using the same Step 7 participant-session cohort, yielding 36 participant-sessions, 30 questions, and 1075 valid trials. The realized block metadata were merged from the cleaned behavioral table, so all diagnostics and models use the actual session order rather than an idealized task template. Descriptively, the valid-trial counts by early blocks began as block 1: 39 abstract trials, block 1: 69 concrete trials, block 2: 39 abstract trials, block 2: 68 concrete trials.

Primary Model

Metric	Value
Included participants	36
Included questions	30
Included trials	1075
Condition beta	3.020e-08
Condition SE	2.276e-08
Condition z statistic	1.327
Condition p-value	0.184407
Condition 95% CI	[-1.440e-08, 7.481e-08]
Global block-order beta	4.425e-09
Within-block-position beta	-6.165e-09
Condition-order beta	3.790e-09
Word-count beta	-4.611e-08
ENEM-correctness beta	3.287e-09
Participant random-intercept variance	7.622e-15
Question random-intercept variance	6.295e-14

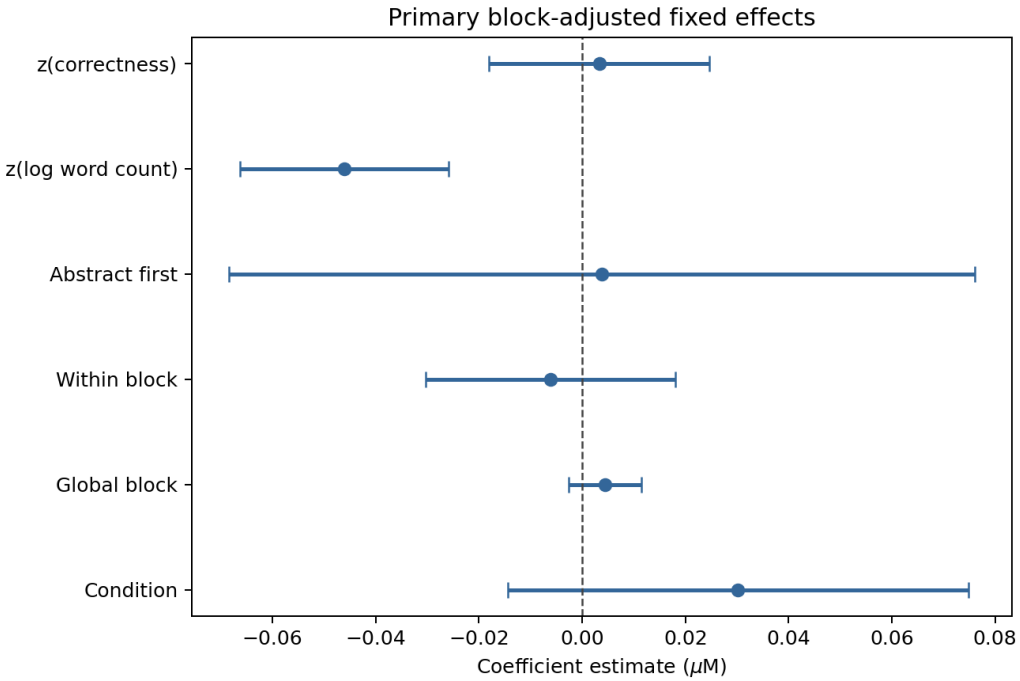

Figure 1: Primary block-adjusted fixed-effect estimates for the front ROI variable-window model.

Descriptive Block Diagnostics

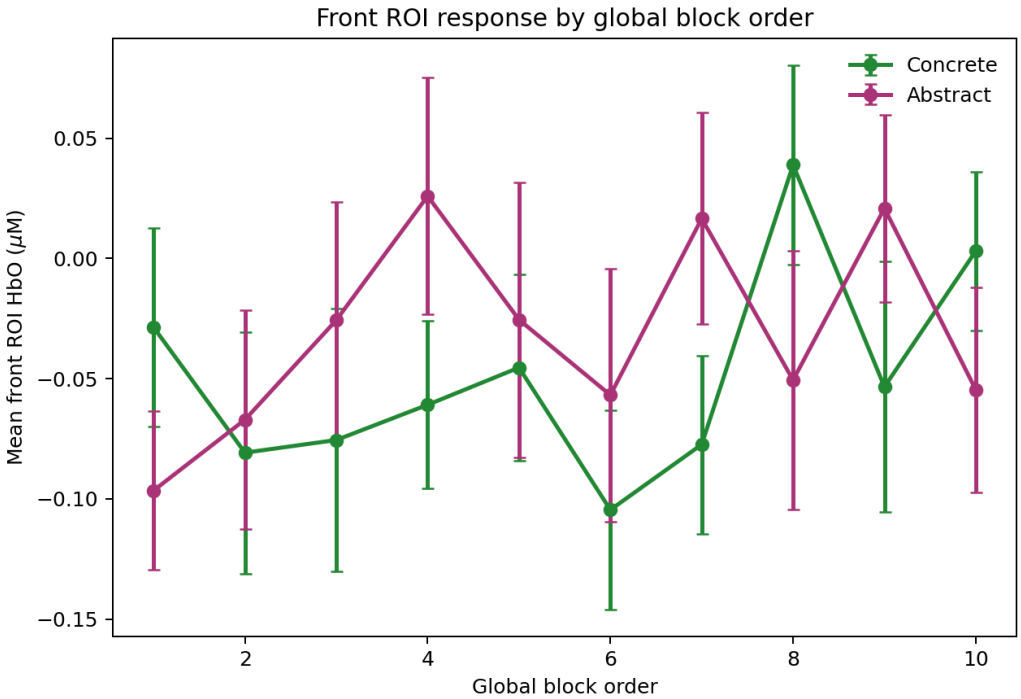

Figure 2: Mean front ROI response by realized global block order and condition.

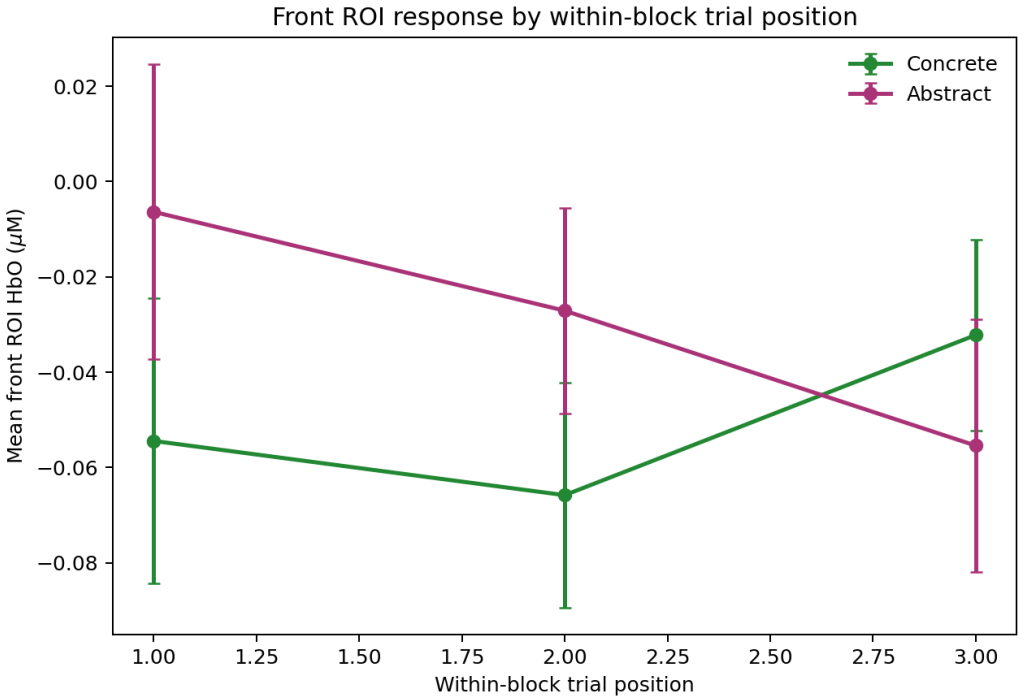

Figure 3: Mean front ROI response by within-block question position and condition.

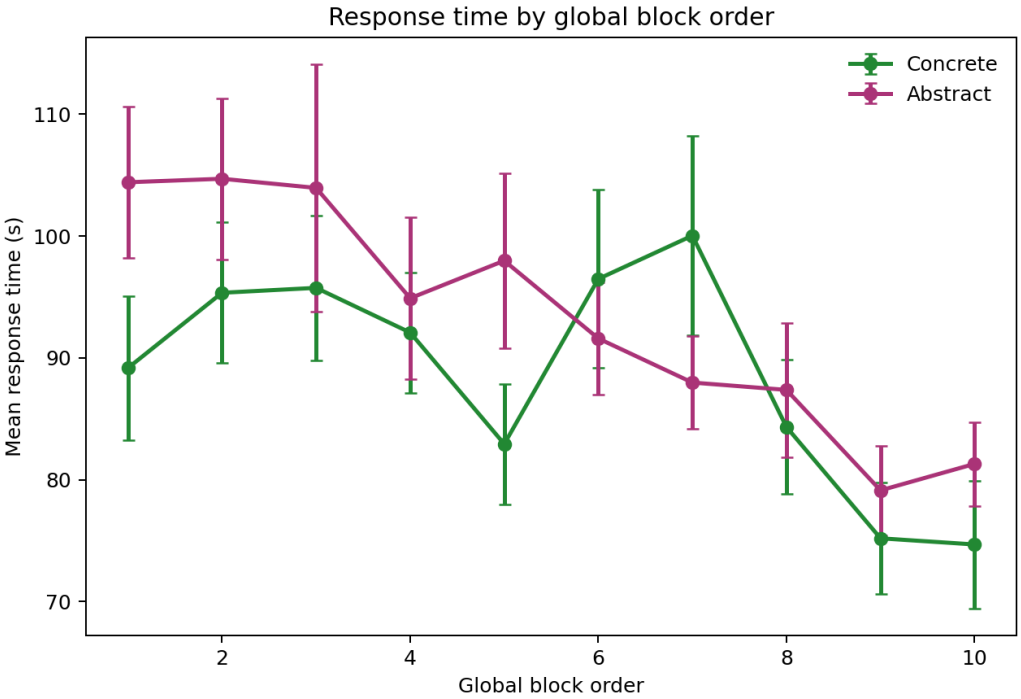

Figure 4: Mean response time by realized global block order and condition.

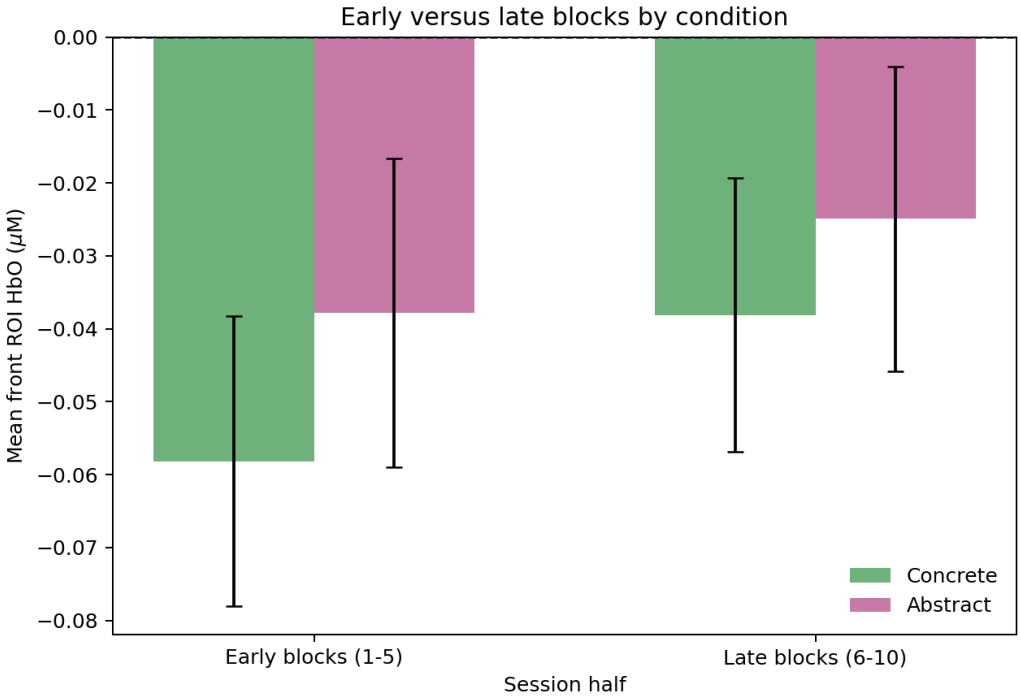

Figure 5: Early-versus-late block-half descriptive comparison by condition.

Global block	Condition	Valid trials
1	Abstract	39
1	Concrete	69
2	Abstract	39
2	Concrete	68
3	Abstract	38
3	Concrete	69
4	Abstract	39
4	Concrete	68
5	Abstract	39
5	Concrete	69
6	Abstract	69
6	Concrete	38
7	Abstract	69
7	Concrete	38
8	Abstract	69
8	Concrete	39
9	Abstract	69
9	Concrete	39
10	Abstract	69
10	Concrete	39

Secondary Models And Benchmarks

Model	Target term	Beta	SE	p	95 Condition by global block order interaction model
Condition x global block order	-9.442e-11	1.424e-08	0.994710	[-2.801e-08, 2.782e-08]	
Condition by within-block position interaction model	Condition x within-block position	-3.125e-08	2.469e-08	0.205681	[-7.965e-08, 1.715e-08]
Condition by condition-order interaction model	Condition x abstract-first order	4.397e-08	8.269e-08	0.594901	[-1.181e-07, 2.060e-07]
Early versus late block-half interaction model	Condition x late-half	-7.685e-09	7.371e-08	0.916963	[-1.522e-07, 1.368e-07]
Secondary back ROI block-adjusted model	Adjusted abstract vs concrete effect	1.391e-08	1.761e-08	0.429482	[-2.060e-08, 4.842e-08]
Fixed-window front ROI benchmark model adjusted for block structure	Adjusted abstract vs concrete effect	2.235e-08	2.052e-08	0.276214	[-1.788e-08, 6.257e-08]

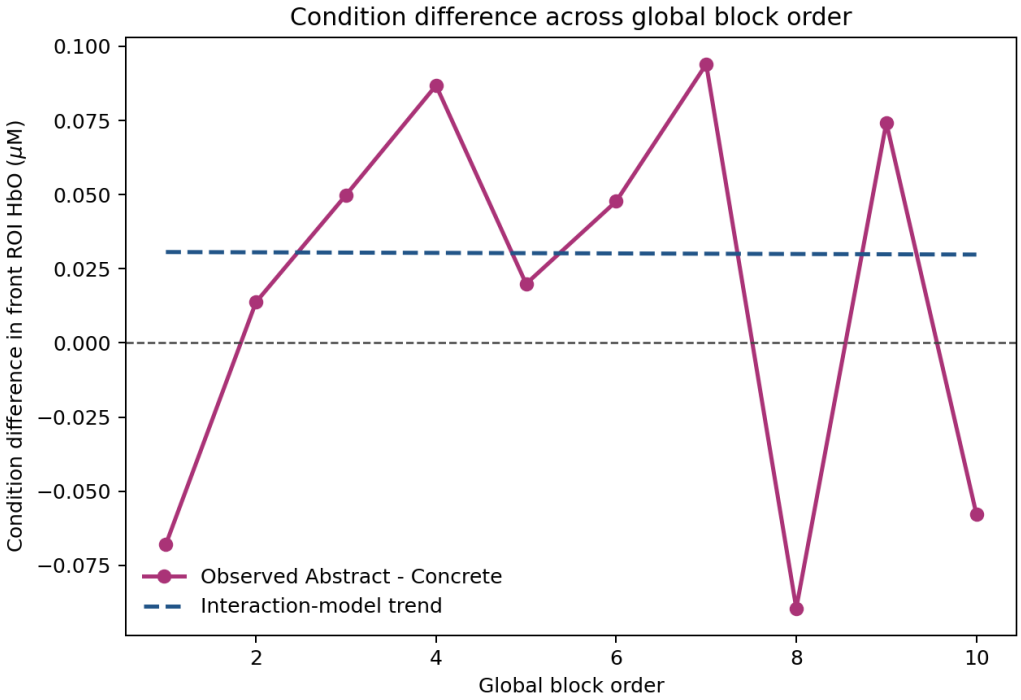

Figure 6: Observed condition difference across global block order with the interaction-model trend overlaid.

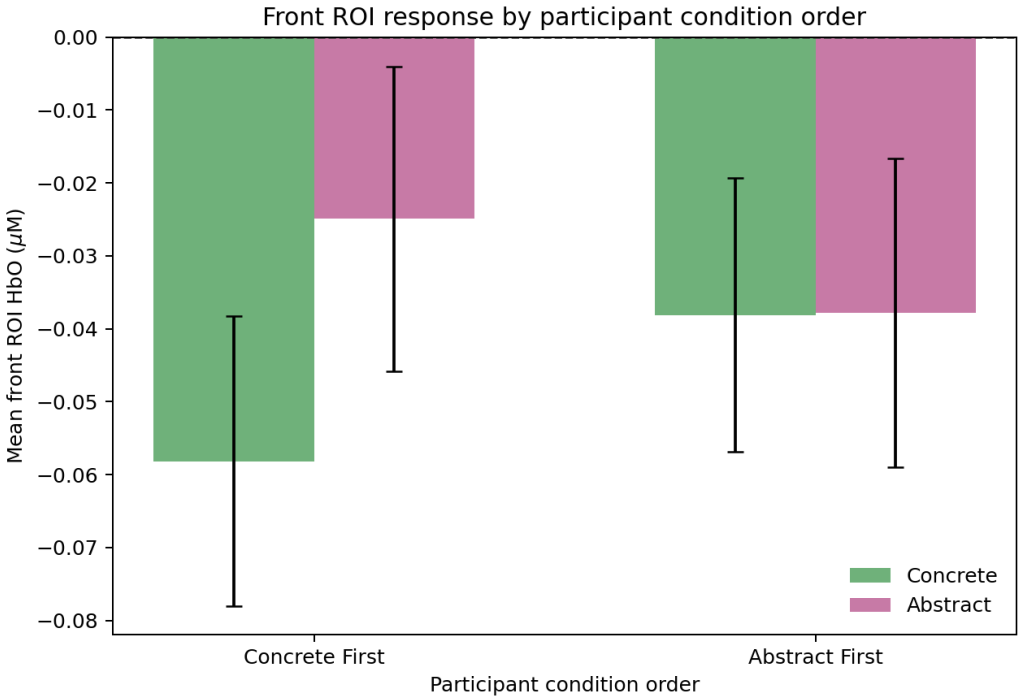

Figure 7: Front ROI response by participant-level condition order.

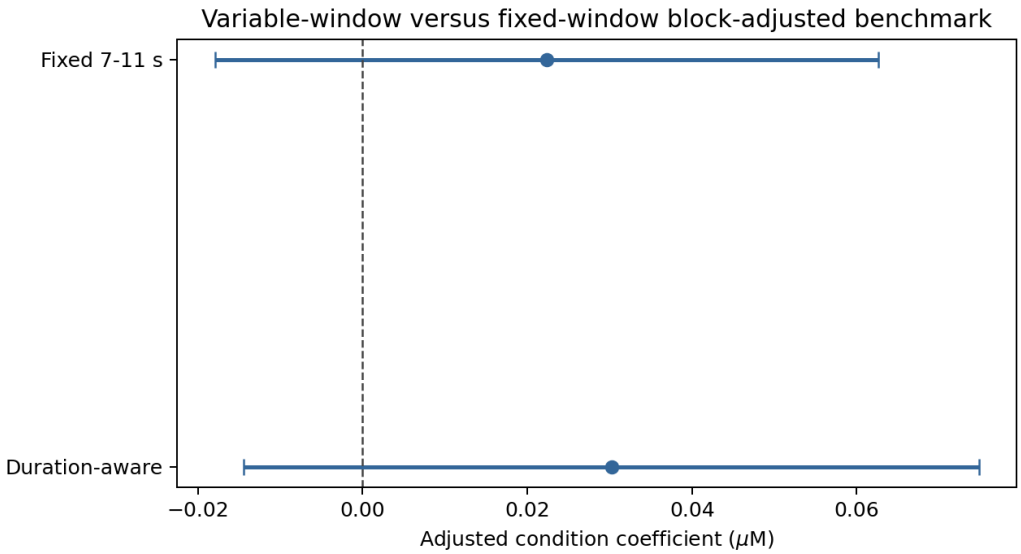

Figure 8: Benchmark comparison between the duration-aware and fixed-window block-adjusted front ROI models.

Warnings And Notes

- `MODEL_PRIMARY_FRONT_BLOCK_ADJUSTED`: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.
- `MODEL_CONDITION_BY_BLOCK_ORDER`: Model type=MixedLM; optimizer=lbfgs; fallback_used=False;

warnings=lbfgs: The MLE may be on the boundary of the parameter space.

- `MODEL_CONDITION_BY_POSITION`: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.
- `MODEL_CONDITION_BY_ORDER_CONDITION`: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.
- `MODEL_EARLY_VS_LATE_SENSITIVITY`: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.
- `MODEL_BACK_BLOCK_ADJUSTED`: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.
- `MODEL_FIXED_WINDOW_BENCHMARK`: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.

Final Conclusion

In the block-adjusted mixed-effects analysis, the abstract-versus-concrete effect in the left frontal ROI was reduced and was not statistically significant after controlling for global block order, within-block trial position, participant condition order, question length, and item difficulty (beta=3.020e-08, SE=2.276e-08, p=0.184407, 95